

MOHAMMED MUDASSIR UDDIN

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RESEARCH SUMMARY

Machine learning researcher and first author of an ICLR 2027 submission on long-context retrieval in hybrid linear-attention language models. Work spans mechanistic interpretability (scalable circuit extraction, released with a public audit of what the code reproduces), efficient LLM inference (task-aware quantization routing for agents) and edge ML (few-shot pruning to 142 ms per image on a Raspberry Pi 4). Currently a Junior Research Fellow on WISER, an Indo-German project mapping pest-infested paddy from RGB, thermal and LiDAR imagery, and a visiting research collaborator at KFUPM on the faithfulness of clinical AI explanations. Builder of LaTeX Intelligence (VS Code Marketplace) and contributor to pgmpy and pyGAM.

EDUCATION

Muffakham Jah College of Engineering and Technology, Osmania University Hyderabad, 2023 – 2026
B.E. Computer Science and Engineering | CGPA 7.99/10

State Board of Technical Education and Training (SBTET), Telangana Hyderabad, 2020 – 2023
Diploma in Computer Science and Engineering | 80%

RESEARCH EXPERIENCE

Junior Research Fellow (JRF) | WISER, Indo-German project, MJCET Hyderabad, 2026 – Present
– WISER is a Government of India and Germany collaboration; under Dr. Tasneem Bano Rehman I detect pest-infested paddy crops from RGB, thermal and LiDAR imagery.

– Processing the multimodal imagery into geotagged outputs (latitude and longitude) that map which field areas are most affected.

Visiting Research Collaborator | King Fahd University of Petroleum and Minerals (KFUPM) May 2026 – Present
– Verifying explanation faithfulness in sepsis clinical decision support with Prof. Hamdi Al-Jamimi: perturb the evidence an explanation names and test whether the recommendation shifts as the explanation predicts.

– Sepsis-3 labels; MIMIC-IV and PhysioNet 2019 first, then HiRID (Swiss ICU) to test whether faithfulness transfers across hospital systems.

Undergraduate Researcher | MJCET, with Dr. Fahmina Taranum 2024 – 2026
– Led the ICLR 2027 study [1]: 49 models from 15 hybrid-architecture designs, one mechanism changed at a time, retrieval tested at every depth up to $64\times$ the training length.
– Co-developed two Google TPU Research Awards proposals: TPU-native evidence memory and KV-cache control for long-horizon agents; perturbation-grounded genome modelling in JAX.

Undergraduate Researcher | MJCET, with Dr. Tasneem Bano Rehman 2025 – 2026
– First author of [3]: disease-aware channel scoring inside a prune, meta-learn, prune pipeline for few-shot plant disease detection on edge hardware.

PUBLICATIONS

[1] **Layer Order and Key Binding Decide When Hybrid Models Learn to Retrieve** M. M. Uddin, F. Taranum, T. B. Rehman, S. Siddiqui, S. Rizwan, S. Abdus Salam, Q. Nayeem, S. U. Begum, S. Kauser, M. Shareef, A. R. Ahmed (*Under review, ICLR 2027*)

Global-first block with bound keys: all **11/11** runs learned retrieval by step **300** and kept **98.0%** recall at **$64\times$** the training length; ablating that one layer drops recall from 99.0% to 0.2%.

[2] **Hierarchical Sparse Circuit Extraction from Billion-Parameter Language Models through Scalable Attribution Graph Decomposition** M. M. Uddin, S. Alam, M. K. Pasha (*arXiv:2601.12879; under review, Discover Artificial Intelligence*)

Coarse-to-fine circuit search, $O(n^2 \log n)$ worst case, **0.09 s** over 413 features; released with a five-seed verification suite and a public claim audit of what the code reproduces.

[3] **Meta-Learning Guided Pruning for Few-Shot Plant Pathology on Edge Devices** M. M. Uddin, S. Alam, M. K. Pasha, T. B. Rehman, F. Taranum, A. Begum (*arXiv:2601.02353; under review, JAIR*)
ResNet-18 from 11.2M to 2.5M parameters (**78%** smaller), **142 ms** per image (**7 FPS**) on a Raspberry Pi 4, **+3.8** points over Meta-Prune at equal size.

[4] **Small Object Detection in Drone Aerial Imagery with LAF-YOLOv10** Q. Nayeem, F. Taranum, M. M. Uddin (*arXiv:2609.14560*)

Composability stress test: four validated modules combined reach **24.0%** mAP@0.5 on VisDrone against 31.8% for YOLOv10n; the deficit traces to one interaction.

[5] **AgentCompress: Task-Aware Compression for Affordable Large Language Model Agents** Z. A. K. Taha, M. M. Uddin, S. Alam (*arXiv:2601.05191*)

Per-step FP16/INT8/INT4/INT2 routing for LLM agents; the paper reports **68.3%** less compute at **96.2%** task success on 290 research workflows.

PROFESSIONAL EXPERIENCE

- Founder** | Auragenzy Hyderabad, Jan 2025 – Feb 2026
– Built and led the agent orchestration and evaluation stack for an AI SaaS for no-code workflow automation; paying users within months of launch.
- Co-founder** | Brevenia Remote, Mar 2023 – Aug 2024
– Social media growth agency started without funding; ran campaigns for 15+ clients with a four-person content team.
- Machine Learning Engineer (Intern)** | RentEra Remote, Dec 2022 – May 2023
– Refined production ML models on real-estate rental data through feature engineering and error analysis, reducing live inference error.
- Junior ML Engineer (Apprentice)** | Techieyan Technologies Hyderabad, May 2022 – Nov 2022
– Ran structured exploratory analysis and statistically validated experiments that surfaced the feature signals behind product decisions.

PROJECTS AND OPEN SOURCE

- LaTeX Intelligence** | TypeScript, VS Code API 2026
– LaTeX-to-Word converter with native OMML equations and live cross-reference fields, plus plain-language diagnosis of failed builds; published on the VS Code Marketplace; 397 tests, 153 KB, no runtime dependencies.
- Federated Data Node** | Python, SQLite, Avro 2026
– Offline-first collection node for Telugu, Tamil, Hindi and Bengali text in which raw text never leaves the device, enforced by four independent egress guards; runs on a 4 GB Raspberry Pi 4 over 2G; 188 tests.
- AlphaEarth** | Python, Google Earth Engine, gradient-boosted trees 2026
– Parcel-level flood and wildfire damage from two Sentinel-2 passes, scored against independent damage records; 24,608 parcels, leave-one-event-out validation, flood macro-F1 0.906.
- Guardian-AI** | Python, Streamlit, MITRE ATLAS 2025 – 2026
– Prompt-injection filter for LLMs that read security logs: 34 rules re-run on decoded, homograph-folded and re-joined inputs; caught 12/12 obfuscated attacks on a regression set at 1.1 ms median latency; 113 tests.
- Vak** | Sarvam AI speech, translation and LLM APIs 2026 – Present
– Open-source AI literacy and interpretability platform built natively in Telugu and Hindi; accepted to the Sarvam AI Startup Program.
- Open-source contributions** | pgmpy, pyGAM 2026
– 11 pull requests to pgmpy (d-separation, inference and model-utility fixes) and 4 to pyGAM (p-values via Wood’s rank-*r* eigendecomposition, link-function overflow, scikit-learn estimator contract).

HONORS AND ACHIEVEMENTS

- **Polaris Fellowship (2026)**: one of ten fellows selected nationally.
- **Erdős Institute Research Cohort (2026)** and **Sarvam AI Startup Program (2026)**, the latter for Vak.
- **Research proposals under review (2026), PI Dr. Fahmina Taranum**: Indian Council of Social Science Research (ICSSR) minor research project (INR 15 lakh), Google TPU Research Awards (AetherState; CausalGenome-JAX), Amazon Research Awards (temporal belief-state auditing for multi-agent systems), GARA as Co-PI (circuit-level cost of locally-quantum decoders).
- **Agentathon 2025**: participant in the event that set the Guinness World Record for the largest agentic AI hackathon (2,300+ developers).

LICENSES & CERTIFICATIONS

Neo4j Certified Professional (2026) | Neo4j & Generative AI (2026) | Claude Certified Architect, Foundations, Anthropic (2026, 944/1000) | CS50x, Harvard (2026) | Neural Networks and Deep Learning, DeepLearning.AI (2023) | IBM Data Science Professional Certificate (2023)

TECHNICAL SKILLS

Research Areas Mechanistic interpretability, long-context architectures, LLM efficiency (quantization, pruning), few-shot learning, multimodal remote sensing (RGB, thermal, LiDAR), explanation faithfulness, causal inference

Languages Python, C++, TypeScript, Java, SQL

ML and Systems PyTorch, JAX, TensorFlow, TFLite, Hugging Face Transformers, vLLM, bitsandbytes, GPTQ, LangGraph, Neo4j, Docker, GCP, AWS, Git